

Michael Chen

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EDUCATION

Master of Data Analytics Specializing in AI (Co-op)

Expected 08-2026

Western University, London, Canada

- CGPA: **4.0/4.0**
- Relevant Coursework: Artificial Intelligence, Reinforcement Learning, Unstructured Data, General linear Models, Deep Learning

HBSc in Mathematics & Computer Science

09-2020 to 09-2024

University of Toronto SG, Toronto, Canada

- Relevant Coursework: Neural Networks, **Machine Learning**, Image Understanding, Data Analysis

PROFESSIONAL EXPERIENCE

AI Developer | Dumpling Collectables, Toronto, Canada

01-2022 to 07-2025 (3 yr 7 mos)

- Achieved **90%** accuracy in automated quality assessment reducing evaluation time by **98%** from 5 minutes to 10 seconds by developing Python OpenCV classification system with custom image preprocessing pipeline
- Generated **15 hours monthly** operational efficiency gains by designing **n8n**-based automation workflow integrating web scraping, data processing, and real-time alerting to identify market opportunities
- Optimized decision-making processes increasing ROI by **200%** by leveraging predictive analytics and historical data modeling for data-driven risk assessment and inventory optimization

AI Data Analyst | Golf Town, Burnaby, Canada

02-2025 to 07-2025 (6 mos)

- Reduced discovery time by **98%** from **30 min to 30 sec** and increased service capacity by **3x** by deploying **RAG architecture** with **LangChain** and vector database autonomously handling customer inquiries
- Built SKU-level grounding solution converting raw inventory into natural language descriptions enabling AI to suggest technical alternatives for out-of-stock items based on launch angle and spin specifications
- Implemented recursive feedback loop optimizing prompts and retrieval methods based on floor consultation performance improving recommendation accuracy and preventing incorrect suggestions

PROJECTS

Comparative RAG System: GraphRAG vs LightRAG Evaluation | GraphRAG, LightRAG, RAGAS, Neo4j, AWS 02-2026

- Implemented both Microsoft **GraphRAG** and **LightRAG** frameworks with custom optimization for entity extraction, relationship mapping, and community detection; leveraged **AWS SageMaker** for model hosting and **Bedrock** for LLM API access, handling **500K+** document chunks on **AWS EC2** and **S3** infrastructure
- Key findings: **LightRAG** demonstrated **3–5x** faster indexing speed and **50–70%** lower operational costs while maintaining comparable answer quality (comprehensiveness score: 8.2 vs 8.4) across domain-specific datasets (legal, medical)

CRM Integration Platform | n8n, REST API, Docker, JavaScript

11-2025

- Eliminated **25+ hours** of manual data entry weekly processing **10K+ contacts** by deploying self-hosted **n8n** platform with **Docker** Compose using webhook and HTTP Request nodes synchronizing Fibery and HubSpot APIs
- Built containerized ETL pipeline implementing Smart Merge algorithm with **n8n's** JavaScript code nodes for field-level comparison achieving **100%** automation of cross-platform synchronization processing **500+ daily** API calls
- Reduced operational costs by **75%** from **\$800 to \$200/month** by migrating from SaaS to self-hosted Docker solution with custom data transformation logic

Hybrid Multi-Agent RAG Framework | LangGraph, LlamaIndex, Python

09-2025

- Achieved enterprise-grade RAGAS scores (faithfulness **0.95**, answer relevancy **0.92**, context precision **0.89**) by developing production-ready **multi-agent system** with **LangGraph** orchestration and **LlamaIndex RAG**, improving answer correctness from 0.60 to **0.95**
- Optimized RAG pipeline performance achieving **35%** improvement through dimensional embedding tuning, hybrid BM25+dense retrieval, and advanced prompt engineering with role-based prompts and structured outputs
- Engineered stateful conversation framework with contextual memory, conversation history tracking, and hierarchical knowledge sharing enabling agents to maintain context across multi-turn interactions

SKILLS

- **Programming Languages:** Python, JavaScript/TypeScript, SQL, Java
- **AI/ML Technologies:** PyTorch, scikit-learn, OpenCV, LangChain, LlamaIndex, LangGraph, LLMs, RAG/Agents
- **Infrastructure & DevOps:** AWS (Lambda, S3, EC2, SageMaker, Bedrock), Docker, CI/CD, Git/GitHub, Kubernetes, Azure
- **Tools & Frameworks:** PostgreSQL, Neo4j, MongoDB, React, REST APIs, n8n, Power BI